

Ramesh Sirimalla

Forward Deployed Engineer — Real-Time Data Streaming & Generative AI

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SUMMARY

Forward Deployed Engineer with 15+ years of experience shipping production-grade real-time data platforms and Generative AI in healthcare and financial services. At CVS Health, built multi-agent AI systems (LangGraph, RAG, MCP) that **cut compliance losses by \$10–15M a year, handle 200,000–500,000 patient replies a day, and contributed to a \$700M yearly increase in messaging revenue**, while **cutting the deployment cycle from 12 weeks to 8**. Works directly with business stakeholders and partners from discovery through production, backed by deep Kafka and streaming expertise.

CORE SKILLS

Streaming & Messaging: Confluent Kafka, Kafka Streams, Kafka Connect, KSQL, Apache Flink, Redpanda, Azure Service Bus, Spark Streaming, StreamSets

Generative & Agentic AI: LLMs, Prompt Engineering, RAG, LangChain, LangGraph, OpenAI API, Amazon Bedrock, Model Context Protocol, LLM Evaluation, Guardrails, FAISS, Pinecone, Chroma

Programming & Frameworks: Java, Python, SQL, C++, Spring Boot, Spring Integration, Spring Batch, Hibernate, PySpark, Pandas

Data, Cloud & DevOps: Oracle, SQL Server, MySQL, MongoDB, Redis, Google Spanner, Hadoop, AWS, Docker, Kubernetes (AKS, GKE), Jenkins, GitHub Actions, Splunk, Grafana

EXPERIENCE

Forward Deployed Engineer

Dec 2023 — Present

CVS Health — Dallas, TX

Agentic AI for Patient Outreach Compliance (TCPA & Do Not Contact)

- Cut CVS Health's legal compliance losses by an estimated \$10–15 million a year by designing and building a multi-agent AI system with LangGraph that screens patient outreach against the Do Not Contact list and TCPA rules across SMS, email, RCS, and IVR.
- Grounded the agents' compliance decisions with retrieval-augmented generation (RAG) over CVS compliance policies, patient consent records, and TCPA regulations.
- Connected the agents to the federal TCPA and Do Not Call (DNC) lists through the Model Context Protocol (MCP), so every outreach decision is checked against current federal data.
- Owned the solution end to end with business stakeholders, from development through production deployment, delivering real-time compliance alerts and an up-to-date dashboard for the business.

AI-Powered Patient Reply Handling (SMS, IVR & Email)

- Developed a model that uses CVS Health's internal NLP / LLM model to understand and respond to 200,000–500,000 patient replies a day across SMS (CVS's production short code), IVR, and email.
- Detects the intent of every reply and responds automatically: opt-outs (STOP), opt-ins (JOIN), UNDO, REFILL, HELP, YES/NO confirmations for refills and appointments, and free-text patient questions.

Multi-Agent Outreach Analytics for Pharmacy Business Teams

- Built a multi-agent system with LangGraph and RAG that gives business stakeholders across Pharmacy Personalization (RxP), Pharmacy Connect (RxC), IDW, and CRDW up-to-date Grafana and Splunk dashboards on patient outreach across SMS, IVR, email, RCS, and push.

- **Increased outreach capacity by 25%**, contributing to a **\$700 million yearly increase in revenue** from CVS Health's messaging platform.

Technical Leadership

- **Cut the deployment cycle from 12 weeks to 8 weeks** (about 33% faster), getting new capabilities to patients and the business sooner.
- Lead engineering teams delivering healthcare solutions with product and business partners, and mentor junior engineers on streaming design, best practices, and technical skills.

Senior Lead Engineer

Jun 2020 — Dec 2023

CVS Health — Dallas, TX

Retail Orchestration Channel Management (ROCM) — Tier 1 Orchestration Hub

- Designed and built Spring Boot services for a **real-time compliance-check API** and a **Patient Preferences update API** serving CVS.com users.
- Built inbound RESTful services for **dispositions from third-party applications**, and integrated internal **submit-order, refill-reminder, and schedule-refill** APIs.
- Built **Spring Integration** applications on the **Confluent Kafka** API to capture closing events in real time, apply business logic, and call REST APIs of other CVS organizations to keep systems in sync.
- Built Spring Boot scheduler applications that load ROCM configuration nightly from **Oracle to MongoDB**.
- Set up **high and critical alerting with C3 and Splunk** for this Tier 1 orchestration hub.

Real-Time Patient & Customer Notifications

- Led high-level design and development of **real-time data streaming applications** that deliver a near-real-time experience to CVS patients and customers across **SMS, email, push, and IVR**.
- Worked directly with **third-party messaging partners** to integrate their SMS, IVR, and email platforms through RESTful web services and take the integrations into production.
- Built **Kafka Streams** applications with Spring Boot that prevent duplicate messages and guarantee delivery within the **legal time window** for patient communications.

Streaming Proofs of Concept

- Built a POC decoupling RESTful services with **Azure Service Bus** for payment-transaction audits during the RxP prescription purchase cycle, with stateful computation in **Apache Flink**.
- Evaluated next-generation streaming platforms through multiple POCs on **Redpanda, Apache Kafka, and Apache Flink** across healthcare use cases.

Engineering Practices

- Led code reviews to maintain quality standards and compliance with healthcare industry regulations.

Senior Data Engineer

Jan 2019 — May 2020

Early Warning Services (Zelle) — Scottsdale, AZ

Real-Time Kafka & Spring Boot Services

- Developed Spring Boot services with **Kafka Connect and KSQL** to store audit transactions for real-time dashboard applications.
- Designed complex **KStream topologies** to dedupe messages and join two or more topics into a final payload published downstream.
- Built Spring Boot applications on Confluent Kafka to keep the inventory system up to date in real time.
- Developed a common **gateway API** with Spring Boot and Redis for load balancing, and integrated third-party APIs.
- Ran a Kafka proof of concept that was presented successfully to the client.

Data Pipelines, Reporting & Monitoring

- Designed scalable pipelines for real-time transaction processing and large-dataset ingestion, with ETL from multiple sources into a central repository.
- Built **StreamSets** pipelines for **Mastercard** reconciliation reports, and Grafana dashboards for **Visa and Zelle** reporting using Spring Boot microservices.
- Built Splunk and Grafana dashboards giving business and management users real-time visibility.

Leadership & Delivery

- Led the development team in designing and delivering solutions, and resolved conflicts between development teams and stakeholders.
- Gathered requirements and defined data architecture with cross-functional teams; managed database security and access-control policies, code reviews, and deployments.

Kafka Lead Developer

Jun 2018 — Dec 2018

CBRE — Dallas, TX

Sequentra Real Estate Data Integration

- Built the Kafka-based integration that streams data from **Sequentra**, CBRE's cloud corporate real estate platform (properties, transactions, projects, occupancy, and space management), to the external **LeaseAccelerator** lease-accounting system through Kafka messages and in-house microservices.
- Implemented complex **Confluent Kafka producer** use cases publishing **3 million records** into the Kafka cluster.
- Built **pub-sub** producer and consumer services for payments, reports, occupancies, and other use cases, updating customer applications in real time.
- Implemented **replay and retry** logic for business needs, with consumers that reset offsets using the Kafka **consumer seek API**.
- Worked closely with **data scientists**, providing the data inputs required by business requirements, and took part in application design document (ADD) discussions with the strategic architect.
- Built a proof of concept provisioning AWS infrastructure with **Terraform**, and worked on **CI/CD** pipelines with DevOps tools.

Big Data Developer

Apr 2015 — May 2018

New York State (ITS) — Albany, NY

Big Data & Real-Time Analytics

- Improved real-time analytics by integrating **Kafka and Spark Streaming** into existing infrastructure.
- Designed big data solutions with **Hadoop** ecosystem tools, with scalable distributed storage and retrieval for large datasets.
- Developed and optimized complex SQL on Oracle, SQL Server, and MySQL.
- Worked with data scientists to refine machine learning models, and with business stakeholders to define new analytics capabilities.

Java Application Development

- Built and maintained scalable Java applications with **Spring and Hibernate**, and integrated third-party APIs.
- Analyzed system performance to remove bottlenecks, and resolved production issues and defects.
- Documented software designs and system architecture, and led code reviews for quality, security, and performance.

Senior Software Engineer

Apr 2013 — Mar 2015

New York State Police (DCJS) — Albany, NY

- Developed, tested, and maintained Java applications, designing solutions and custom algorithms for complex technical challenges.
- Integrated third-party APIs, and investigated and fixed production issues reported by users.

- Estimated project costs from customer requirements and delivered with cross-functional Agile teams.

Senior Software Engineer

Nov 2011 — Mar 2013

Pearson VUE — Iowa City, IA

NBPTS National Board Certification — Fee Processing

- Built the **fee-processing system** for the **National Board for Professional Teaching Standards (NBPTS)** certification program, which certifies about 55,000 teachers and school counselors.
- Gathered requirements with the client and business analysts; wrote the Functional Specification (FSD), System Design Document (SDD), and high- and low-level designs, with UML diagrams in IBM Rational XDE.
- Developed the core **makeAJournalEntry** flow and about **10 use cases** (Distribute Existing Funds, Scholarship Distribution, Approval, Third Party Payer Account Setup, and more), plus journal-entry reports and stored procedures.
- Built **CyberSource** credit-card and eCheck reconciliation reports, and implemented a POST URL to fix Hosted Order Page timeouts.
- Developed with Java, Struts, EJB, DAO, JavaScript, and Ajax; integrated systems with HTTP, XML web services, and **SOAP**.
- Played a major role in taking the system **live and stabilizing it**, then rolled it out to other brands.

EDUCATION & CERTIFICATIONS

Bachelor of Engineering, Computer Science — MVSR Engineering College (Osmania University), Hyderabad, India, 2001 – 2005 (First Class)

Confluent Fundamentals Accreditation — Confluent